A. Clinical ECG sources

PTB-XL



20,487 ECGs

Georgia



8,302 ECGs

Chapman



8,634 ECGs

Ningbo



28,956 ECGs

Harmonised label space

13 diagnostic classes retained across all sources

Leave-one-source-out evaluation

Fold 1	Fold 2	Fold 3	Fold 4
test: **PTB-XL**	test: **Georgia**	test: **Chapman**	test: **Ningbo**

train on three sources, test on the fourth

B. Reference models and deployment-oriented scaling

10 reference models (unconstrained)

published topology, float32, no device budget

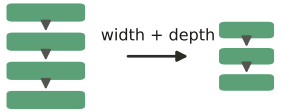
Two scaling strategies for MCU deployment

Uniform width scaling



thinner, same depth

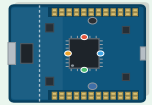
Joint width-depth scaling



thinner and shallower

Training, quantisation, MCU deployment

Train (FP32) → **Quantise** (INT8) → **Deploy on MCU**



measured on STM32F411VET6:
Flash, peak SRAM, latency, macro-AP

C. Evaluation outputs

RQ1. Cross-source generalization

- external discrimination across four held-out sources
- class-specific AP and paired architecture comparisons

RQ2. Scaling and MCU efficiency

- compare scaling strategies under deployment budgets
- assess prediction, quantisation effects, Flash, SRAM, latency

RQ3. Performance-resource trade-offs

- identify favorable deployed configurations
- trade-offs among external macro-AP, latency and memory